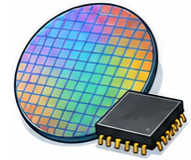


ENGLISH FOR WORK / SEPTEMBER 2026

Semiconductor English



Learner workbook

Work with realistic cases, develop your own response, and use the answer section after you have tried the tasks.

PRACTICE THAT TRANSFERS TO WORK

Read a realistic case. Study complete conversations. Find the right language, speak, get feedback, and try again.

Eight lessons | Intermediate to advanced | Independent or classroom study

For semiconductor process engineers, yield engineers, product engineers, equipment engineers, fab supervisors, quality teams, test and packaging teams, foundry coordinators, supply planners, applications engineers, and technical program managers.

These cases practice communication. Actual equipment, construction, and safety work follows approved procedures and authorized specialist decisions.

All scenarios, figures, and model responses are fictional teaching material. Use invented details in practice.

[Open this course on English Ladder](#)

Your course at a glance

Use the lesson numbers to match this document with the website and the other guides.

01 Wafer Fabrication Flow and Process Integration

Explain a useful distinction in plain English.

02 Lithography, Reticles, and Critical Dimensions

Clarify missing information before responding.

03 Deposition, Etch, CMP, and Process Windows

Separate observations, assumptions, and conclusions.

04 Metrology, SPC, and Yield Learning

Distinguish completed work from pending work and explain its impact.

05 Defect Density and Cleanroom Contamination

Make a request with a clear action, purpose, and realistic timing.

06 Equipment Uptime, Recipes, and Tool Matching

Compare options using the same criteria and state the tradeoff.

07 Packaging, Test, and Reliability Qualification

Separate observations, assumptions, and conclusions.

08 Foundry, Tape-Out, PDK, and Capacity Communication

Negotiate scope or timing while making conditions explicit.

Choose your pace

Quick practice: spend 15 minutes reading one conversation and rehearsing one scenario. Full lesson: allow 45-60 minutes for language work, three conversations, two speaking scenarios, and feedback.

Level support

B1 learners can use the frames and a partner. B2 learners can try a response before reading the model. C1 learners can add the harder follow-up and reduce preparation time. These are teaching suggestions, not certified CEFR level ratings.

After practicing: open the AI prompt workshop to choose a vocabulary, grammar, role-play, or new-dialogue activity.

LESSON 01 / READ AND NOTICE

Wafer Fabrication Flow and Process Integration

Explain a useful distinction in plain English.

SITUATION

A new colleague confuses wafer fabrication with packaging. A process overview needs to explain where the current work takes place.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

wafer - Thin semiconductor substrate, usually silicon, on which integrated circuits are fabricated.

fab - Semiconductor fabrication facility where wafers are processed through manufacturing steps.

process flow - Ordered sequence of semiconductor manufacturing steps, layers, inspections, holds, and decision points.

node - Technology generation or process family, often associated with feature size, performance, density, and design rules.

3. Notice the language

Start with the reader's question. Explain one unfamiliar term with familiar words, then give a relevant example or contrast. Check understanding without asking only 'Do you understand?'.

In this context, ... means

The difference is that

How would you explain that distinction to a colleague?

Compare this wording: We need to operationalize the implementation.

LESSON 01 / PRACTICE AND PRODUCE

Make a complex point clear

4. Check the language

1. Which sentence explains 'backlog' rather than repeating it?
 - A. The backlog is the work completed during the last period.
 - B. The backlog is the maximum work the team can finish in a period.
 - C. The backlog is the work still waiting to be completed.
2. Which follow-up best checks whether your explanation was clear?
 - A. Was the explanation detailed enough for you?
 - B. How would you describe the next step in your own words?
 - C. Would you like me to repeat the explanation?

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 01 / 05 • CONVERSATIONS / 1 OF 3

Our work is in the fab, not packaging

Fictional onboarding: a new manufacturing planner confuses wafer fabrication with packaging and asks where the process-integration team works.

01 • Manufacturing planner: Does our process-integration team attach the finished chips to their packages in this area?

02 • Process engineer: Our current work is wafer fabrication in the fab, before the packaging stage you describe.

03 • Manufacturing planner: Could you explain that distinction without assuming I know every operation in the process flow?

04 • Process engineer: Here we coordinate steps that form device structures on wafers. Packaging prepares separated dies for their intended connections.

05 • Manufacturing planner: So the wafer remains the working piece across many fabrication steps, rather than arriving already packaged.

06 • Process engineer: Yes. For this overview, follow the wafer through our documented flow before moving to downstream packaging.

07 • Manufacturing planner: Where does the node name fit? Is it the name of a packaging operation?

08 • Process engineer: No. Here it identifies the process technology offering; it doesn't describe the package assembly step.

09 • Manufacturing planner: I'll label the overview by fabrication and packaging stages instead of combining them as one activity.

10 • Process engineer: Good. Ask the downstream owner to confirm their boundary before adding packaging details to our flow.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 01 / 05 • CONVERSATIONS / 2 OF 3

A lot traveler is not a circuit drawing

Fictional fab induction: a planner sees a lot traveler and mistakes its operation sequence for the product's circuit layout.

01 • Production planner: This traveler has dozens of steps. Is it showing where every transistor sits on the wafer?

02 • Integration engineer: No. It records the lot's manufacturing route, while the layout describes the designed structures.

03 • Production planner: Then the process flow tells us the order of operations, not the circuit connections themselves.

04 • Integration engineer: Exactly. It helps us follow what happens to the wafer through the fab.

05 • Production planner: Can I assume every product on the same node follows this identical route?

06 • Integration engineer: Don't assume that. Check the approved route for the product and configuration you're planning.

07 • Production planner: That means a familiar node label isn't enough to choose the next operation.

08 • Integration engineer: Correct. Use the lot's controlled manufacturing information rather than filling gaps from another product.

09 • Production planner: I'll ask the route owner to explain the step labels before I summarize the schedule.

10 • Integration engineer: I'll join that discussion and distinguish the manufacturing sequence from the design information you saw earlier.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 01 / 05 • CONVERSATIONS / 3 OF 3

Wafer completion does not mean shipment readiness

Fictional planning meeting: a wafer lot has completed its listed fab operations, but downstream test and packaging status are unknown.

01 • Account planner: The lot has finished its fab operations. Can I tell the customer the packaged devices are ready?

02 • Manufacturing coordinator: Not from that status alone. Fab completion and packaged-product readiness are different points in the flow.

03 • Account planner: Which distinction should I explain without inventing a downstream completion date we don't have?

04 • Manufacturing coordinator: Say the wafer lot completed its listed fabrication operations; downstream test and packaging status still need confirmation.

05 • Account planner: Would describing the wafers as finished chips create the same confusion for a nontechnical customer?

06 • Manufacturing coordinator: It could. Use the actual stage name and explain what work that stage does not include.

07 • Account planner: I'll request the downstream status from its owner before making a shipment-readiness statement.

08 • Manufacturing coordinator: Please keep the product and route identifiers with that request so the correct lot is checked.

09 • Account planner: Understood. Today's update will report fab completion, not completed packaging or an approved shipment.

10 • Manufacturing coordinator: That is accurate and gives the next team a precise question they can answer.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 01 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Wafer Fabrication Flow and Process Integration

A new colleague confuses wafer fabrication with packaging. A process overview needs to explain where the current work takes place.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You are unfamiliar with one technical term in the case. Ask for a plain-English explanation and then paraphrase it. Do not pretend to understand a term you cannot explain.

Second round: Your listener is new to this field. Replace two specialist expressions with clear explanations without changing the meaning.

Model for scenario 1: Wafer fabrication forms devices on the wafer; packaging happens later in the product flow. Our update concerns the fabrication stage. I'll show the relevant process steps before discussing the issue.

Scenario 2 | A process-flow slide skips inspection

A fictional training slide shows deposition followed by etch. The approved route includes an inspection between them. A new planner thinks the slide authorizes skipping inspection. Explain an overview versus an operational route.

Role A: Process-integration engineer: explain the slide's limited teaching purpose.

Role B: Manufacturing planner: ask which controlled record determines the actual sequence.

Possible opening: The slide simplifies the flow for orientation; it does not replace the approved lot route.

Success checks

- Distinguish an overview from controlled manufacturing instructions.
- Avoid authorizing an operational change.
- Agree to confirm the actual route with its owner.

Add a complication: The planner says the shorter slide must be the newer process.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 02 / READ AND NOTICE

Lithography, Reticles, and Critical Dimensions

Clarify missing information before responding.

SITUATION

A lithography review reports a critical-dimension shift but does not identify the layer or measurement conditions.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

lithography - Patterning process that transfers circuit features to a wafer using light, masks, and photoresist.

photoresist - Light-sensitive material used in lithography to define patterns on a wafer.

reticle - Photomask used in lithography to project circuit patterns onto a wafer.

critical dimension - A measured feature size on a wafer that must stay within specification for device performance and yield.

3. Notice the language

Name the exact word, fact, or requirement you need clarified. Ask one focused question at a time. Repeat your understanding and give the other person a chance to correct it.

When you say ..., do you mean ...?

Could you clarify which ...?

So, my understanding is Is that right?

Compare this wording: Could you explain what does this mean?

LESSON 02 / PRACTICE AND PRODUCE

Ask a precise question

4. Check the language

1. Which indirect question has the usual word order?
 - A. Could you confirm when starts the review?
 - B. Could you confirm when the review starts?
 - C. Could you confirm when does the review start?
2. Someone says the report is 'ready.' Which question best clarifies its approval status?
 - A. Has the report been approved, or is it ready for review?
 - B. Can you send the report to me this afternoon?
 - C. Who will present the report at the meeting?

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 02 / 05 • CONVERSATIONS / 1 OF 3

Which layer shows the CD shift?

Fictional lithography review using the lesson case: a critical-dimension shift is reported without a layer or measurement conditions.

01 • Lithography lead: The summary reports a critical-dimension shift. Which layer and measurement conditions does that refer to?

02 • Metrology engineer: Those details aren't in the summary. I'll retrieve the measurement record before interpreting the shift.

03 • Lithography lead: Please include the sampled features and whether the measurements were taken before or after etch.

04 • Metrology engineer: Why does that stage distinction matter for a report labeled simply lithography performance?

05 • Lithography lead: Different measurement stages address different parts of the flow, so the label may overstate attribution.

06 • Metrology engineer: Understood. We also need the relevant reticle and photoresist process identifiers, not just the lot name.

07 • Lithography lead: Yes, where available in the record. Don't fill missing conditions from a previous lot.

08 • Metrology engineer: Should I hold the causal interpretation while I collect that context from the measurement owner?

09 • Lithography lead: Exactly. We can acknowledge the reported shift without claiming which operation caused it.

10 • Metrology engineer: I'll return with the identified layer, measurement stage, and any remaining gaps clearly separated.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 02 / 05 • CONVERSATIONS / 2 OF 3

The reticle revision is missing

Fictional pattern review: two wafer images are compared, but only one image has a reticle revision recorded.

01 • Process engineer: These two images look different. Were they produced with the same reticle revision?

02 • Lithography engineer: Only the first image includes that identifier. We shouldn't assume the second used the same revision.

03 • Process engineer: Could you confirm the revision from the exposure record before we compare feature dimensions?

04 • Lithography engineer: Yes. I'll also check that the images show the same layer and comparable measurement locations.

05 • Process engineer: A colleague wants to call the difference a photoresist issue. Is there supporting evidence yet?

06 • Lithography engineer: Not yet. We first need to establish that the compared patterns and conditions are actually comparable.

07 • Process engineer: Then our immediate question is traceability, not which process parameter to change.

08 • Lithography engineer: Correct. The reticle identifier connects the observed pattern to the design information used for exposure.

09 • Process engineer: I'll label the comparison incomplete and keep both original image records for the review.

10 • Lithography engineer: I'll obtain the missing identifiers and explain which conclusions remain open after that check.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 02 / 05 • CONVERSATIONS / 3 OF 3

A wafer average without measurement locations

Fictional metrology discussion: a report gives one average critical dimension for a wafer but omits the measurement-site map.

01 • Integration lead: The average critical dimension is unchanged. Does that mean every measured location stayed consistent?

02 • Metrology engineer: The average alone doesn't tell us that. Could we review the site-level results and sampling map?

03 • Integration lead: The map isn't attached. Which details should I request from the measurement team first?

04 • Metrology engineer: Ask where the sites were, which features were measured, and which method produced the values.

05 • Integration lead: Should I also confirm that the reference wafer used the same measurement conditions?

06 • Metrology engineer: Yes. Otherwise similar averages may come from comparisons with different measurement boundaries.

07 • Integration lead: I won't describe the full wafer as uniform until the spatial evidence has been reviewed.

08 • Metrology engineer: Good. That preserves the average as an observation without making it answer a different question.

09 • Integration lead: I'll request the missing map and ask the owner to explain any unmatched sites.

10 • Metrology engineer: Then we can discuss the critical-dimension pattern with the context needed for a useful review.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 02 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Lithography, Reticles, and Critical Dimensions

A lithography review reports a critical-dimension shift but does not identify the layer or measurement conditions.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You need to know exactly what information is missing. Give one answer only after your partner asks a focused question; then ask them to confirm their understanding.

Second round: The other person uses an ambiguous word such as 'ready', 'approved', or 'urgent'. Clarify it before continuing.

Model for scenario 1: Which layer and measurement conditions does the shift refer to? Please include the target, observed result, and reference recipe so we can discuss the same comparison.

Scenario 2 | Nanometers or a relative percentage?

A fictional lithography slide labels a shift as 3 without a unit. The underlying measurement table has not been retrieved. Clarify whether this is a dimensional value, percentage, or other measure before interpreting it.

Role A: Process reviewer: request the unit, reference, layer, and measurement stage.

Role B: Report author: acknowledge the missing label and commit to checking the source.

Possible opening: Could you clarify what the value three measures and which reference it is compared with?

Success checks

- Do not infer the missing unit.
- Request the measurement context and reference.
- Keep significance and cause unconfirmed until the source is checked.

Add a complication: A manager asks you to assume nanometers because the meeting concerns critical dimensions.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 03 / READ AND NOTICE

Deposition, Etch, CMP, and Process Windows

Separate observations, assumptions, and conclusions.

SITUATION

A process change improves one measured outcome but has not been checked across the proposed operating range.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

deposition - Process of adding material layers to a wafer by physical, chemical, epitaxial, or atomic-layer methods.

etch - Process that removes selected material from a wafer using wet chemistry or plasma-based methods.

CMP - Chemical mechanical planarization; a process that smooths wafer surfaces for later manufacturing steps.

process window - Range of process conditions under which results meet specification with acceptable margin.

3. Notice the language

Say what the evidence shows, identify what is still unknown, and limit the conclusion to that evidence. Words such as 'may', 'suggests', and 'in this sample' are useful when the uncertainty is real. Do not soften an urgent, verified safety concern.

The available evidence shows

This may indicate ..., but

We have not yet established whether

Compare this wording: The pilot proves this will work everywhere.

LESSON 03 / PRACTICE AND PRODUCE

Match certainty to evidence

4. Check the language

1. Which sentence preserves uncertainty about an unconfirmed cause?
 - A. The change may have contributed to the delay.
 - B. The change must have caused the delay.
 - C. The change definitely caused the delay.
2. A result comes from a small pilot in one location. Which phrase accurately limits the claim?
 - A. In similar locations, we can now expect the same result ...
 - B. Across the service, the observed result was ...
 - C. In this pilot, the observed result was ...

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 03 / 05 • CONVERSATIONS / 1 OF 3

A better center point is not a process window

Fictional process review using the lesson case: a change improves one measured outcome at one setting but has not been checked across the proposed range.

01 • Process engineer: The changed setting improved our measured result. Can we say the process window is wider?

02 • Integration lead: We only tested one setting. A wider operating range hasn't been demonstrated by that observation.

03 • Process engineer: So the improvement is real for the measured case, but the range claim is still an assumption.

04 • Integration lead: Exactly. We need evidence across the proposed conditions before describing an expanded process window.

05 • Process engineer: Should we also check effects on downstream etch instead of looking only at deposition output?

06 • Integration lead: Yes, through the approved evaluation plan. Improving one stage doesn't establish the integrated outcome.

07 • Process engineer: I'll present the measured result separately from the hypothesis about a broader operating range.

08 • Integration lead: Please retain the tested conditions and identify the untested range explicitly in the review.

09 • Process engineer: I'll ask the evaluation owner which additional comparisons are needed before recommending any process change.

10 • Integration lead: Good. Today's conclusion can remain useful without turning one point into a qualified window.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 03 / 05 • CONVERSATIONS / 2 OF 3

A smoother surface with an open thickness question

Fictional CMP evaluation: one polished sample shows lower measured roughness, but remaining-film thickness has not been assessed.

01 • CMP engineer: The sample has lower measured roughness after polishing. I'd like to call the new condition better.

02 • Integration engineer: Better on roughness, yes. Have we assessed the remaining film thickness under that condition?

03 • CMP engineer: Not yet. Chemical mechanical polishing, or CMP, was evaluated only for this surface measurement so far.

04 • Integration engineer: Then the integrated result is incomplete. A surface improvement doesn't answer the open thickness question.

05 • CMP engineer: Should I describe thickness loss as a confirmed problem in the review to emphasize the risk?

06 • Integration engineer: No. It is unmeasured here, not established as excessive. Keep observation and concern distinct.

07 • CMP engineer: I'll request thickness evidence from the evaluation owner before proposing a wider process window.

08 • Integration engineer: Please include comparable conditions so the new measurement can be interpreted alongside the roughness result.

09 • CMP engineer: The slide will say improved measured roughness in one sample, with thickness assessment pending.

10 • Integration engineer: That gives us a bounded finding and a clear next question for the process review.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 03 / 05 • CONVERSATIONS / 3 OF 3

An etch result after two changes

Fictional engineering trial: both the incoming deposited film and the etch condition changed; a profile measurement improved.

01 • Etch engineer: The profile improved in this trial. I think the changed etch condition caused the improvement.

02 • Deposition engineer: The incoming film also changed, so this trial doesn't isolate the etch condition's contribution.

03 • Etch engineer: You're right. We altered two parts of the process flow before measuring the result.

04 • Deposition engineer: We can report the combined trial outcome while keeping the individual explanations as hypotheses.

05 • Etch engineer: Would repeating the same combined condition establish which of the two changes mattered?

06 • Deposition engineer: It could test repeatability of the combination, but it wouldn't separate the two effects by itself.

07 • Etch engineer: Then let's ask the evaluation owner for a comparison that addresses those effects under controlled conditions.

08 • Deposition engineer: Agreed. We should preserve the trial records and the incoming-film description for that planning discussion.

09 • Etch engineer: I'll remove the single-cause claim and avoid saying the process window has been qualified.

10 • Deposition engineer: Good. The review can recognize progress while clearly stating what the evidence does not yet isolate.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 03 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Deposition, Etch, CMP, and Process Windows

A process change improves one measured outcome but has not been checked across the proposed operating range.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You need to tell a colleague what is known. Ask which part is confirmed and which part is an assumption. Challenge one statement that sounds more certain than the case supports.

Second round: Someone asks for a yes-or-no answer when the evidence supports only a qualified answer. Be concise while preserving the uncertainty.

Model for scenario 1: The result is encouraging at the tested settings. We have not established the wider process window. I'll distinguish the observed improvement from the conditions that still need evaluation.

Scenario 2 | The edge sites are still unmeasured

A fictional deposition trial meets its target at center-wafer sites. Edge sites were not measured, and no full-wafer claim has been assessed. Discuss the result and a request for further evidence without changing a recipe.

Role A: Deposition engineer: state that measured center sites meet the target and identify the unmeasured edge sites.

Role B: Integration reviewer: ask what remains unknown about spatial performance.

Possible opening: The measured center sites meet the target, but the edge sites are still unmeasured.

Success checks

- Limit the finding to measured sites.
- Distinguish missing evidence from a failed result.
- Agree an evaluation question for the responsible owner.

Add a complication: A colleague wants to title the slide Full-wafer improvement confirmed.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 04 / READ AND NOTICE

Metrology, SPC, and Yield Learning

Distinguish completed work from pending work and explain its impact.

SITUATION

A pilot lot has 92% yield compared with 90% in a reference lot. The sample sizes and product mix differ.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

metrology - Measurement discipline used to verify process, dimension, film, defect, and device characteristics.

SPC - Statistical process control; use of control charts and limits to monitor process stability.

yield - Share of wafers, die, units, or lots that meet requirements after manufacturing, test, or qualification.

excursion - Manufacturing event or trend outside expected control limits, specifications, or normal process behavior.

3. Notice the language

Lead with the current state, then the important change or open item. Use completed-action language only for completed work. Add a next update point when it is known; do not invent a date to make the message sound complete.

We have completed ...; ... remains open.

As of ..., the status is

The next update will cover

Compare this wording: Everything is progressing well.

LESSON 04 / PRACTICE AND PRODUCE

Give a useful status update

4. Check the language

1. Which sentence says a review is finished?
 - A. The team is completing the review.
 - B. The team plans to complete the review.
 - C. The team has completed the review.
2. Which update distinguishes a plan from a confirmed fact?
 - A. The delivery plan proves that arrival will be Friday.
 - B. Delivery is planned for Friday; the carrier has not confirmed it.
 - C. Delivery is confirmed for Friday because we planned it.

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 04 / 05 • CONVERSATIONS / 1 OF 3

Ninety-two percent is not yet a matched improvement

Fictional yield review using the lesson case: a pilot lot has 92% yield versus 90% for a reference lot; sample sizes and product mix differ.

01 • Yield analyst: The pilot yield is ninety-two percent, compared with ninety percent in the reference lot.

02 • Integration manager: Can I report a confirmed process improvement in the executive summary this afternoon?

03 • Yield analyst: Not yet. Sample sizes and product mix differ, so the current comparison isn't matched.

04 • Integration manager: What work is complete, and what still prevents a clear statement about improvement?

05 • Yield analyst: The reported lot yields are available. A like-for-like comparison and uncertainty review remain pending.

06 • Integration manager: Please don't bury the good result, but show why the two-point difference needs interpretation.

07 • Yield analyst: I'll display both rates with their denominators and product breakdown once those records are verified.

08 • Integration manager: Does the metrology review offer an independent explanation for the difference at this stage?

09 • Yield analyst: No confirmed explanation yet. I'll keep measurement findings separate from the yield attribution.

10 • Integration manager: Good. Report the observed rates and the next analysis step, not an improvement already established.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 04 / 05 • CONVERSATIONS / 2 OF 3

An SPC signal awaits investigation

Fictional monitoring handoff: an SPC chart flags one point for review; the measurement record is available, but the process investigation has not begun.

01 • Metrology engineer: The statistical process control chart flagged one point, and the underlying measurement record is available.

02 • Shift manager: Does that establish a product failure, or does it identify something that needs investigation?

03 • Metrology engineer: It identifies a signal under this chart's configured rules. Product disposition hasn't been established here.

04 • Shift manager: Then which parts of the excursion review are complete and which remain pending?

05 • Metrology engineer: The signal was captured. Verification of measurement context and the process review are still pending.

06 • Shift manager: Please hand those open items to their owners rather than describing the entire review as complete.

07 • Metrology engineer: I'll attach the chart version and measurement context through the established quality workflow.

08 • Shift manager: Can you state the operational impact without inventing a hold instruction or assuming release authority?

09 • Metrology engineer: I'll say disposition awaits the responsible team's review and refer any operational decision to them.

10 • Shift manager: That is the right boundary. Keep the observed signal visible while the investigation proceeds.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 04 / 05 • CONVERSATIONS / 3 OF 3

A measurement rerun is complete, the yield analysis is not

Fictional yield-learning project: a corrected metrology report is ready, but its relationship to final test yield has not been analyzed.

01 • Program lead: The metrology rerun is finished. Can we mark the yield-learning action complete as well?

02 • Yield engineer: Only the measurement step is complete. We haven't analyzed its relationship to final test yield.

03 • Program lead: I thought collecting corrected measurements automatically resolved the explanation for the yield change.

04 • Yield engineer: It improves the evidence, but the interpretation still requires matched lot and test information.

05 • Program lead: Which dependency should the status report show so the remaining work has a clear owner?

06 • Yield engineer: The test-data owner needs to confirm the corresponding lot records before we perform that comparison.

07 • Program lead: Does that mean the corrected metrology results should stay hidden until the whole analysis finishes?

08 • Yield engineer: No. Show them as completed evidence, with the yield relationship explicitly unassessed.

09 • Program lead: I'll separate measurement completion from analysis completion and request the matched records at the next checkpoint.

10 • Yield engineer: I'll confirm the scope of the comparison once the records arrive and keep unresolved attribution visible.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 04 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Metrology, SPC, and Yield Learning

A pilot lot has 92% yield compared with 90% in a reference lot. The sample sizes and product mix differ.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You need to brief someone who has only one minute. Ask what is complete, what is open, and which open item affects the next action.

Second round: The listener asks you to reduce the update to two sentences. Preserve the most important completed and pending items.

Model for scenario 1: The pilot's reported yield is two percentage points higher. The lots differ in size and mix, so the comparison needs qualification. I'll show those details alongside the headline result.

Scenario 2 | The denominator excludes untested dies

A fictional pilot report lists 96 passes out of 100 tested dies. Twenty additional dies have not been tested. A colleague calls the whole group 96% passing. Explain the completed test result and pending coverage.

Role A: Yield analyst: state the tested denominator and the untested count.

Role B: Program manager: ask what can be said about all 120 dies.

Possible opening: The ninety-six percent rate applies to the hundred tested dies, not all one hundred twenty.

Success checks

- Preserve the tested denominator.
- Distinguish untested from passing.
- Agree a follow-up for missing test coverage.

Add a complication: The manager asks to count untested dies as passes because no failures were recorded.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 05 / READ AND NOTICE

Defect Density and Cleanroom Contamination

Make a request with a clear action, purpose, and realistic timing.

SITUATION

A contamination review needs the timing and location of observed particles. The current note says only "cleanroom problem."

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

defect density - Number or rate of defects on a wafer, die, layer, lot, or process area.

particle - Small contaminant that can create defects, yield loss, reliability risk, or process instability.

cleanroom - Controlled manufacturing environment designed to limit particles, humidity, electrostatic risk, and contamination.

contamination control - Practices used to prevent particles, residues, metals, organics, moisture, or handling errors from affecting wafers or devices.

3. Notice the language

State the action and the information needed. Explain why it matters and specify a deadline only when one is given or agreed. 'Could you' is polite; repeated apologies can obscure a legitimate request. Urgent situations may require a direct request.

Could you provide ... so that ...?

Please confirm ... by [agreed time].

If that timing is not possible, please let me know

Compare this wording: Please revert soonest with the necessary.

LESSON 05 / PRACTICE AND PRODUCE

Ask for an actionable next step

4. Check the language

1. What does 'Please send it by Thursday' normally mean?
 - A. Keep sending it regularly until Thursday.
 - B. Send it no later than Thursday.
 - C. Send it on Thursday, but not earlier.
2. Which request gives the recipient enough detail to act?
 - A. Could you confirm the document version needed for the review?
 - B. Could you confirm that the review is important?
 - C. Could you confirm that you received my earlier message?

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 05 / 05 • CONVERSATIONS / 1 OF 3

Replace cleanroom problem with location and time

Fictional contamination review using the lesson case: a note says only cleanroom problem and lacks particle timing and location.

01 • Contamination engineer: The note says cleanroom problem. Could you provide the time and location of the observed particles?

02 • Shift coordinator: I can ask the reporting technician, but the note doesn't identify a tool or sampling point.

03 • Contamination engineer: Please request those details and the observation method before our review, or identify what cannot be recovered.

04 • Shift coordinator: Should I also ask the technician to name the contamination source they suspect?

05 • Contamination engineer: They can label a hypothesis, but the immediate priority is the observed evidence, not a guessed source.

06 • Shift coordinator: Understood. Would defect density help, or do we need the inspected area to interpret it?

07 • Contamination engineer: We need the relevant count and area information rather than an unexplained density value.

08 • Shift coordinator: I'll request the original record through our contamination-control channel and preserve the observation wording.

09 • Contamination engineer: Thank you. Avoid circulating unsupported blame about a person, tool, or cleaning activity.

10 • Shift coordinator: I'll return with verified details and explicit gaps so the review can decide its next steps.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 05 / 05 • CONVERSATIONS / 2 OF 3

More particles on a larger inspected area

Fictional inspection comparison: one record lists 12 particles over twice the inspected area of a record listing 8; methods have not been compared.

01 • Quality coordinator: The new inspection found twelve particles instead of eight. Can we call contamination worse?

02 • Defect engineer: The inspected area doubled, so raw counts alone don't establish that comparison. Please obtain both area records.

03 • Quality coordinator: Should I calculate defect density immediately and use that as the final conclusion?

04 • Defect engineer: First confirm comparable methods and what each count includes. A ratio doesn't repair unmatched definitions.

05 • Quality coordinator: What precise request should I send the inspection owner for our afternoon review?

06 • Defect engineer: Ask for inspected area, counting criteria, method, and sampling locations for both records.

07 • Quality coordinator: I'll also ask which entries are verified so we don't quietly fill missing information ourselves.

08 • Defect engineer: Good. Keep the particle counts visible but label the comparative interpretation as pending.

09 • Quality coordinator: Can we agree to review the source records before discussing a contamination-control response?

10 • Defect engineer: Yes. The responsible review team needs comparable evidence before drawing conclusions or proposing operational changes.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 05 / 05 • CONVERSATIONS / 3 OF 3

A photo without an observation timestamp

Fictional cleanroom review: a particle image is attached to an issue, but the upload date is the only recorded time.

01 • Process coordinator: The particle image was uploaded at nine. Can I say the observation occurred then?

02 • Contamination reviewer: No. Upload time and observation time can differ. Please request the original capture record.

03 • Process coordinator: The image also lacks a sampling-location label. Should I infer the location from the issue owner?

04 • Contamination reviewer: Please don't. Ask the reporting owner to identify the actual location and any missing metadata.

05 • Process coordinator: I'll request capture time, location, and method before tomorrow's contamination review, with gaps explicitly reported.

06 • Contamination reviewer: That gives the request a clear purpose and avoids attaching false precision to the timeline.

07 • Process coordinator: Should we delay every discussion until the metadata is complete, even though the image exists?

08 • Contamination reviewer: We can discuss the documented observation and missing context, but not attribute it to a particular event.

09 • Process coordinator: I'll preserve the image and distinguish upload time from the unconfirmed observation time in the handoff.

10 • Contamination reviewer: Good. That helps the review team assess the evidence without an invented sequence of events.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 05 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Defect Density and Cleanroom Contamination

A contamination review needs the timing and location of observed particles. The current note says only "cleanroom problem."

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You can help but need a clear request and its purpose. Ask what is required and whether the timing is fixed or negotiable.

Second round: The other person cannot meet the requested timing. Clarify an alternative without inventing authority to change a formal deadline.

Model for scenario 1: Could you provide the observation time, location, and relevant records? A precise description will help the team investigate without assuming a contamination source before reviewing the evidence.

Scenario 2 | Before or after the cleaning event?

A fictional particle report and a cleaning record share a date, but neither includes a usable time. No causal link has been established. Request evidence that could clarify the sequence without issuing contamination-control instructions.

Role A: Contamination reviewer: ask for original observation and cleaning records.

Role B: Shift coordinator: identify the missing sequence and contact the record owners.

Possible opening: The records share a date, but we cannot yet tell whether the observation preceded the cleaning.

Success checks

- Avoid inferring sequence from a shared date.
- Request specific original timing evidence.
- Keep source attribution unconfirmed.

Add a complication: A colleague wants to describe cleaning as the confirmed contamination source.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 06 / READ AND NOTICE

Equipment Uptime, Recipes, and Tool Matching

Compare options using the same criteria and state the tradeoff.

SITUATION

Two tools run the same nominal recipe but show different measured results. A team member assumes their performance is interchangeable.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

tool uptime - Percentage of time equipment is available and qualified for production use.

preventive maintenance - Planned equipment service performed to reduce unplanned downtime, drift, contamination, safety risk, or tool instability.

recipe - Controlled equipment parameters used to run a process step on a wafer, lot, or tool.

tool matching - Effort to make similar manufacturing tools produce equivalent results within defined limits.

3. Notice the language

Use the same time period, scope, and measurement basis for both options. Explain a difference with 'whereas' or 'while'. State what improves and what is given up; do not hide the weaker side of a recommendation.

Both options ..., but

Option A ..., whereas option B

The comparison depends on

Compare this wording: This option is more better.

LESSON 06 / PRACTICE AND PRODUCE

Explain a fair comparison

4. Check the language

1. Which sentence expresses a clear contrast?
 - A. Option A is faster, whereas option B costs less.
 - B. Option A is faster because option B costs less.
 - C. Option A is faster, so option B costs less.
2. A rate increases from 6% to 7%. What is the absolute difference?
 - A. An increase of one percent relative to the original rate.
 - B. An increase of seven percentage points.
 - C. An increase of one percentage point.

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 06 / 05 • CONVERSATIONS / 1 OF 3

The same recipe name does not establish tool matching

Fictional equipment comparison using the lesson case: two tools run the same nominal recipe but produce different measured results.

01 • Production planner: Both tools run the same nominal recipe. Can we treat their output as interchangeable?

02 • Equipment engineer: The measured results differ, so the shared recipe name doesn't establish tool matching.

03 • Production planner: What should we compare before choosing which tool to allocate to the next evaluation?

04 • Equipment engineer: Use comparable product, measurement conditions, configuration records, and relevant maintenance history for both tools.

05 • Production planner: Tool A has better uptime. Would that alone make it the stronger option?

06 • Equipment engineer: It helps availability, but uptime doesn't answer whether the process results meet the same requirements.

07 • Production planner: So the tradeoff may involve availability and demonstrated process performance, not a single winner.

08 • Equipment engineer: Exactly. We should also distinguish measured differences from any unverified explanation about their cause.

09 • Production planner: I'll request a like-for-like comparison and avoid assigning work based only on the recipe label.

10 • Equipment engineer: I'll bring the tool-matching evidence and open questions to the responsible allocation review.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 06 / 05 • CONVERSATIONS / 2 OF 3

Uptime measured with different exclusions

Fictional equipment report: one tool's uptime excludes preventive-maintenance hours, while another report includes those hours in its denominator.

01 • Operations manager: Tool A's uptime is higher than tool B's. Should we prioritize A for capacity planning?

02 • Equipment analyst: Their reports treat preventive maintenance differently, so the percentages aren't directly comparable yet.

03 • Operations manager: Can you explain the mismatch without getting lost in the dashboard calculation details?

04 • Equipment analyst: One percentage excludes planned maintenance time; the other includes it in the time being measured.

05 • Operations manager: Then we need the same measurement boundary before ranking their availability from those figures.

06 • Equipment analyst: Yes. I'll recalculate using an agreed definition and show planned maintenance separately.

07 • Operations manager: Will that establish which tool produces better process results or meets matching requirements?

08 • Equipment analyst: No. Availability and process performance are separate criteria, and both matter to the allocation discussion.

09 • Operations manager: Please bring the aligned uptime view together with the existing process-performance evidence.

10 • Equipment analyst: I will, with unknown matching status visible rather than concealed behind a favorable availability figure.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 06 / 05 • CONVERSATIONS / 3 OF 3

A maintained tool versus an unreviewed backup

Fictional scheduling review: tool A has completed preventive maintenance but awaits post-maintenance review; backup tool B is available but matching status is unknown.

01 • Scheduling lead: Tool A's maintenance is finished. Should we use it immediately instead of the available backup?

02 • Equipment engineer: Maintenance completion is recorded, but the post-maintenance review is pending. Availability alone doesn't establish readiness.

03 • Scheduling lead: Tool B is available now. Does its backup designation mean tool matching is already confirmed?

04 • Equipment engineer: Not in the records we have. Its matching status is unknown and needs owner confirmation.

05 • Scheduling lead: Then both choices have open conditions, although their reasons are different.

06 • Equipment engineer: Correct. Compare readiness evidence, scheduling availability, and product suitability on the same basis.

07 • Scheduling lead: Can we propose A subject to review while requesting B's matching record as an alternative?

08 • Equipment engineer: We can take those conditional options to the authorized planning owner without directing an unapproved run.

09 • Scheduling lead: I'll keep maintenance complete, readiness pending, and backup matching unknown as separate status entries.

10 • Equipment engineer: That makes the tradeoff transparent and avoids treating either tool as qualified by assumption.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 06 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Equipment Uptime, Recipes, and Tool Matching

Two tools run the same nominal recipe but show different measured results. A team member assumes their performance is interchangeable.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You favor one option but have not checked whether the scope is comparable. Ask about one difference, then explain the tradeoff you are willing to accept.

Second round: Your listener focuses only on the lowest price or fastest date. Explain one other relevant comparison criterion.

Model for scenario 1: The recipe name is the same, but the observed results differ. Let's compare the controlled settings and qualification evidence before describing the tools as matched for this process.

Scenario 2 | Identical recipe names, different revisions

Two fictional tools display recipe name R7, but one record lists revision 2 and the other revision 3. Output comparison has not been performed. Discuss what must match before claiming interchangeability.

Role A: Equipment engineer: explain the identifier difference.

Role B: Production planner: ask for comparable configuration and performance evidence.

Possible opening: The displayed recipe names match, but their revision records differ, so interchangeability remains unconfirmed.

Success checks

- Separate recipe name from revision identity.
- Avoid treating newer as automatically qualified.
- Agree a records and performance review before allocation.

Add a complication: A colleague says the newer revision must be suitable for every tool.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 07 / READ AND NOTICE

Packaging, Test, and Reliability Qualification

Separate observations, assumptions, and conclusions.

SITUATION

A package passes one reliability test while two planned tests remain open. A customer slide calls qualification complete.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

package - Protective and electrical interface that connects a semiconductor die to a board or system.

binning - Sorting tested semiconductor units into performance, power, speed, or quality categories.

burn-in - Stress testing used to screen for early-life failures before product release or shipment.

qualification - Evidence-based approval that a process, product, tool, package, supplier, or change meets defined requirements.

3. Notice the language

Say what the evidence shows, identify what is still unknown, and limit the conclusion to that evidence. Words such as 'may', 'suggests', and 'in this sample' are useful when the uncertainty is real. Do not soften an urgent, verified safety concern.

The available evidence shows

This may indicate ..., but

We have not yet established whether

Compare this wording: The pilot proves this will work everywhere.

LESSON 07 / PRACTICE AND PRODUCE

Match certainty to evidence

4. Check the language

1. Which sentence preserves uncertainty about an unconfirmed cause?
 - A. The change must have caused the delay.
 - B. The change definitely caused the delay.
 - C. The change may have contributed to the delay.
2. A result comes from a small pilot in one location. Which phrase accurately limits the claim?
 - A. Across the service, the observed result was ...
 - B. In this pilot, the observed result was ...
 - C. In similar locations, we can now expect the same result ...

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 07 / 05 • CONVERSATIONS / 1 OF 3

One reliability pass with two tests open

Fictional package review using the lesson case: one reliability test passed, two planned tests remain open, and a slide calls qualification complete.

01 • Customer quality lead: The slide says qualification complete. Have all planned package reliability tests actually finished?

02 • Package engineer: No. One test passed, while two remain open. The completion statement is too broad.

03 • Customer quality lead: Can we preserve the positive result without implying the remaining tests have passed?

04 • Package engineer: Yes. State which test passed and list the two pending tests under their current status.

05 • Customer quality lead: Does an open test mean a failure was found, or simply that its outcome isn't available?

06 • Package engineer: Here it means the planned work is still open; no outcome should be invented.

07 • Customer quality lead: Please also keep electrical binning results separate from the overall reliability qualification claim.

08 • Package engineer: Agreed. Classification by test results doesn't replace evidence for every planned reliability condition.

09 • Customer quality lead: I'll revise the customer wording and ask the qualification owner to confirm the next update.

10 • Package engineer: I'll provide the verified test status and avoid a completion date that hasn't been confirmed.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 07 / 05 • CONVERSATIONS / 2 OF 3

A burn-in result is not every reliability result

Fictional device review: the recorded burn-in test passed, but the rest of the qualification matrix has not been reviewed in this meeting.

01 • Program manager: The burn-in report passed. Can I summarize that as all reliability requirements being satisfied?

02 • Test engineer: No. That report covers its specified test conditions, not the entire qualification matrix.

03 • Program manager: Could you explain the distinction for colleagues who see burn-in as a complete reliability guarantee?

04 • Test engineer: It is one defined stress-screening or test activity in this program, not proof about every future condition.

05 • Program manager: We haven't reviewed the other reports here, so their status remains unknown to this discussion.

06 • Test engineer: Correct. We should ask the qualification owner for the complete matrix instead of inferring completion.

07 • Program manager: Should I say the remaining tests are failures to prevent anyone assuming they passed?

08 • Test engineer: No. Unknown status is neither a pass nor a failure. Use the actual evidence boundary.

09 • Program manager: I'll report the burn-in result specifically and leave the overall qualification statement pending confirmation.

10 • Test engineer: I'll request the matrix and keep the package configuration attached to the evidence we discuss.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 07 / 05 • CONVERSATIONS / 3 OF 3

Different bins do not explain the cause

Fictional final-test review: a package lot has more units in a lower performance bin than a reference lot; test-condition comparability is unverified.

01 • Product engineer: More units entered the lower performance bin. Does that prove packaging damaged the devices?

02 • Test analyst: The binning distribution changed, but that observation alone doesn't establish the cause.

03 • Product engineer: What context do we need before interpreting the difference between these two lots?

04 • Test analyst: Comparable test conditions, product configuration, and bin definitions, plus the relevant lot history.

05 • Product engineer: Those checks are pending. I should not treat the lower bin as proof of a packaging defect.

06 • Test analyst: Correct. A bin records classification under the test criteria; attribution needs additional evidence.

07 • Product engineer: Can we still flag the distribution for the package and test teams to review jointly?

08 • Test analyst: Yes. Report the observed counts and preserve the unverified conditions as explicit questions.

09 • Product engineer: I'll avoid calling the lot unqualified solely from this comparison without the program's review.

10 • Test analyst: I'll bring the bin definitions and available records so the responsible team can assess the finding.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 07 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Packaging, Test, and Reliability Qualification

A package passes one reliability test while two planned tests remain open. A customer slide calls qualification complete.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You need to tell a colleague what is known. Ask which part is confirmed and which part is an assumption. Challenge one statement that sounds more certain than the case supports.

Second round: Someone asks for a yes-or-no answer when the evidence supports only a qualified answer. Be concise while preserving the uncertainty.

Model for scenario 1: One test is complete; two are still open. I'll report the results by test and avoid describing the overall qualification as complete until the responsible team confirms the required evidence.

Scenario 2 | The report covers another package configuration

A fictional qualification report names package configuration P1. The proposed customer part uses P2. The relationship between configurations has not been reviewed. Discuss applicability without assuming either equivalence or failure.

Role A: Package engineer: identify the configuration mismatch.

Role B: Customer quality coordinator: ask who can review whether evidence applies.

Possible opening: The report covers P1; we still need an applicability review for the proposed P2 configuration.

Success checks

- Name the evidence boundary.
- Keep equivalence unconfirmed rather than claiming failure.
- Route applicability to the qualification owner.

Add a complication: A colleague says similar external dimensions prove equivalent qualification.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

LESSON 08 / READ AND NOTICE

Foundry, Tape-Out, PDK, and Capacity Communication

Negotiate scope or timing while making conditions explicit.

SITUATION

A customer requests an earlier tape-out slot. Design checks are still open, and foundry capacity is limited.

1. Understand the situation

Identify two confirmed facts, one missing detail, and the person who needs your response. Do not fill gaps with invented facts.

2. Find the words

foundry - Semiconductor manufacturer that fabricates chips for external customers or design companies.

tape-out - Final release of a chip design to the foundry for mask generation and fabrication.

PDK - Process design kit; foundry-provided design rules, models, and files used to design chips for a process.

capacity allocation - Decision process for assigning limited foundry, tool, test, or assembly capacity across products or customers.

3. Notice the language

Acknowledge the goal and explain the available choices. Connect an offer to its actual conditions. A useful response makes room for discussion without promising approval or resources you do not control.

We can ..., provided that

Would you prefer ... or ...?

I can take that proposal for review; I cannot yet confirm

Compare this wording: If you will approve the scope, we can assess the schedule.

LESSON 08 / PRACTICE AND PRODUCE

Offer a workable tradeoff

4. Check the language

1. Which sentence presents a normal future condition clearly?
 - A. If the scope changes, we reviewed the schedule.
 - B. If the scope changes, we will review the schedule.
 - C. If the scope will change, we will review the schedule.
2. Which response negotiates a new request without silently promising extra work?
 - A. We can assess the addition and explain its effect on the agreed scope.
 - B. We have included it, although its scope has not been reviewed.
 - C. We guarantee the original date without assessing the addition.

05 • Conversations

The next three pages contain original fictional conversations for this lesson. Each numbered line is one speaking turn. Read the setting before assigning roles.

LESSON 08 / 05 • CONVERSATIONS / 1 OF 3

An earlier tape-out slot remains conditional

Fictional foundry negotiation using the lesson case: a customer requests an earlier tape-out slot, design checks are open, and capacity is limited.

01 • Customer program manager: Can the foundry move our tape-out earlier? The project team wants a firmer customer date.

02 • Foundry liaison: We can discuss options, but design checks remain open and capacity allocation is limited.

03 • Customer program manager: Could you reserve the earlier slot now and assume the checks will finish in time?

04 • Foundry liaison: I can't promise that. We need the design-readiness position and the foundry's actual allocation decision.

05 • Customer program manager: What can you offer today that helps us plan without creating a false commitment?

06 • Foundry liaison: I can request available options and state the readiness conditions attached to each proposal.

07 • Customer program manager: Please include the required process design kit, or PDK, version in that readiness discussion.

08 • Foundry liaison: Yes. The design team should confirm its check basis rather than assume versions are interchangeable.

09 • Customer program manager: I'll report the earlier date as requested, not secured, until those conditions and capacity are confirmed.

10 • Foundry liaison: I'll return with the foundry response and keep any unresolved checks explicit in the negotiation.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 08 / 05 • CONVERSATIONS / 2 OF 3

A new PDK version before submission

Fictional design handoff: a team has checks against PDK version 6, while a proposed slot refers to version 7; applicability is unconfirmed.

01 • Design manager: Our checks passed against PDK six. Can we accept the proposed slot referring to PDK seven?

02 • Foundry interface engineer: First we need to confirm the applicable submission requirements and the effect of that version difference.

03 • Design manager: Could we simply rename the package so it matches the newer version in the slot note?

04 • Foundry interface engineer: No. A label change would not establish that the design was checked against the required basis.

05 • Design manager: Then what conditional proposal can we take back while preserving the current completed work?

06 • Foundry interface engineer: Ask whether the existing basis is acceptable or what reviewed updates are needed before submission.

07 • Design manager: I'll have the design-check owner assess that question without promising an immediate rerun completion.

08 • Foundry interface engineer: I'll clarify the foundry's slot conditions and whether timing alternatives are available.

09 • Design manager: Until then, our tape-out date remains provisional, with the version question clearly stated.

10 • Foundry interface engineer: Agreed. We'll negotiate from verified readiness and capacity information, not from a renamed file.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 08 / 05 • CONVERSATIONS / 3 OF 3

Splitting a request does not create capacity

Fictional capacity discussion: a customer proposes splitting one manufacturing request into two smaller requests; available foundry capacity has not been confirmed.

01 • Customer planner: Could we split our request into two smaller allocations and secure an earlier start that way?

02 • Foundry account lead: We can ask about phased allocation, but splitting the request doesn't itself create available capacity.

03 • Customer planner: I understand. The design team would still need to meet the same agreed readiness conditions.

04 • Foundry account lead: Yes, and the foundry must confirm whether the proposed phasing fits its actual planning constraints.

05 • Customer planner: Could you present the phased option alongside the original request without canceling the original proposal?

06 • Foundry account lead: I'll clarify how alternatives should be submitted so we don't unintentionally create duplicate commitments.

07 • Customer planner: What should I tell our internal team about timing while that discussion is ongoing?

08 • Foundry account lead: Say phased capacity is under discussion, with no earlier slot confirmed and readiness still subject to review.

09 • Customer planner: I'll keep both options conditional and ask the design owner for an updated check status.

10 • Foundry account lead: I'll bring back the capacity response so we can compare workable choices rather than assumed dates.

Notice, then rehearse

Name each speaker's purpose. Identify one useful expression and the next step or unresolved question. Read the roles aloud, then switch roles.

LESSON 08 / 06 • SAY IT

Two scenarios to practice

Use only the supplied facts. Prepare for two minutes, speak for one minute, then respond to a follow-up. Switch roles and repeat. For solo study, speak both roles aloud.

Scenario 1 | Foundry, Tape-Out, PDK, and Capacity Communication

A customer requests an earlier tape-out slot. Design checks are still open, and foundry capacity is limited.

Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You want the preferred outcome but can accept one tradeoff. Ask for two options, then state which condition you can accept and which still needs approval.

Second round: The preferred option is unavailable. Offer an alternative and clearly state any approval still needed.

Model for scenario 1: We can review the earlier slot, but the open design checks and capacity need confirmation. Could we agree on a decision point that reflects both readiness and the foundry's available schedule?

Scenario 2 | A slot offer expires before a readiness review

A fictional foundry offers a provisional slot with a response requested Thursday. The design-readiness review is Friday, and open checks remain. The liaison cannot approve capacity or waive checks. Negotiate a clarification or timing alternative.

Role A: Customer program manager: explain the mismatch and request an option.

Role B: Foundry liaison: offer to ask about an extension or later slot without guaranteeing either.

Possible opening: The requested response precedes our readiness review, so we cannot yet confirm that the design meets the slot conditions.

Success checks

- Keep open checks visible.
- Offer conditional alternatives within your authority.
- Agree who will confirm readiness and capacity separately.

Add a complication: The customer asks you to describe the open checks as administrative details.

Check: facts accurate | meaning clear | tone appropriate | next action or question explicit

Continue scenario 1 with an AI partner.

Answer workshop

Try the tasks first. The model is one acceptable response to scenario 1, not a script to memorize. Assess your spoken version using the four criteria at the end.

01 | Wafer Fabrication Flow and Process Integration

Editing example: We need to put the plan into practice.

The revised sentence names the action in familiar words. Specialist vocabulary is useful when it adds precision, but repeating abstract nouns can hide the meaning.

Check 1: C - The backlog is the work still waiting to be completed.

A: Completed work is different from work still waiting.

B: This describes capacity, not the work waiting to be completed.

C: This gives the meaning in familiar words.

Check 2: B - How would you describe the next step in your own words?

A: This invites an opinion on detail, not a demonstration of understanding.

B: An open request lets the listener show their understanding.

C: Offering repetition may help, but it does not show how the listener understood the next step.

Model response: Wafer fabrication forms devices on the wafer; packaging happens later in the product flow. Our update concerns the fabrication stage. I'll show the relevant process steps before discussing the issue.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

02 | Lithography, Reticles, and Critical Dimensions

Editing example: Could you explain what this means?

After an introductory phrase such as 'Could you explain', the embedded question uses statement word order: "what this means".

Check 1: B - Could you confirm when the review starts?

A: Place the subject 'the review' before 'starts'.

B: The embedded question uses subject then verb: "the review starts".

C: Remove 'does' and use statement word order after 'confirm'.

Check 2: A - Has the report been approved, or is it ready for review?

A: This question distinguishes two meanings of 'ready' that affect the next action.

B: This asks about delivery timing, not whether approval has happened.

C: This asks about the presenter, not the report's approval status.

Model response: Which layer and measurement conditions does the shift refer to? Please include the target, observed result, and reference recipe so we can discuss the same comparison.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

03 | Deposition, Etch, CMP, and Process Windows

Editing example: The pilot worked in the tested setting; performance elsewhere remains untested.

The revision states the finding and its scope. It avoids turning limited evidence into a universal claim.

Check 1: A - The change may have contributed to the delay.

A: 'May have contributed' presents a possibility without claiming the cause is established.

B: 'Must have' expresses a strong inference that the evidence may not support.

C: 'Definitely' presents the cause as certain.

Check 2: C - In this pilot, the observed result was ...

A: Similarity alone does not establish that the result will transfer.

B: A pilot in one location does not establish a service-wide observation.

C: This locates the finding in the setting actually studied.

Model response: The result is encouraging at the tested settings. We have not established the wider process window. I'll distinguish the observed improvement from the conditions that still need evaluation.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

04 | Metrology, SPC, and Yield Learning

Editing example: Three checks are complete; two remain open and are being reviewed.

The revision gives observable progress. A positive label alone does not tell the reader what is complete or still needed.

Check 1: C - The team has completed the review.

A: 'Is completing' presents the work as in progress.

B: 'Plans to complete' describes an intention, not completion.

C: 'Has completed' states that the action is finished.

Check 2: B - Delivery is planned for Friday; the carrier has not confirmed it.

A: A plan cannot prove a future arrival time.

B: This explicitly labels the date as planned and identifies the missing confirmation.

C: A plan does not establish carrier confirmation.

Model response: The pilot's reported yield is two percentage points higher. The lots differ in size and mix, so the comparison needs qualification. I'll show those details alongside the headline result.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

05 | Defect Density and Cleanroom Contamination

Editing example: Could you send the missing document and confirm when it will be available?

The revision names the document and requested response. It avoids expressions that may be unfamiliar or ambiguous for an international audience.

Check 1: B - Send it no later than Thursday.

A: This interprets a deadline as a continuing activity, which would use "until".

B: 'By' sets the latest point, while 'after' refers to a later time.

C: "By Thursday" permits earlier delivery; it is not restricted to that day.

Check 2: A - Could you confirm the document version needed for the review?

A: The requested action and information are explicit.

B: This checks importance rather than identifying the needed document version.

C: This checks receipt, not which version is required for the review.

Model response: Could you provide the observation time, location, and relevant records? A precise description will help the team investigate without assuming a contamination source before reviewing the evidence.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

06 | Equipment Uptime, Recipes, and Tool Matching

Editing example: This option is better on cost, but slower to deliver.

'Better' already has comparative meaning, so it does not need 'more'. Naming the criterion and tradeoff makes the comparison useful.

Check 1: A - Option A is faster, whereas option B costs less.

A: 'Whereas' marks a contrast between two options.

B: 'Because' claims a causal connection that is not given.

C: 'So' implies one fact causes the other.

Check 2: C - An increase of one percentage point.

A: A one-percent relative increase on 6% would be 6.06%, not 7%.

B: The new rate is 7%; the difference between the rates is one percentage point.

C: Subtracting 6% from 7% gives one percentage point. The relative increase is about 16.7%.

Model response: The recipe name is the same, but the observed results differ. Let's compare the controlled settings and qualification evidence before describing the tools as matched for this process.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

07 | Packaging, Test, and Reliability Qualification

Editing example: The pilot worked in the tested setting; performance elsewhere remains untested.

The revision states the finding and its scope. It avoids turning limited evidence into a universal claim.

Check 1: C - The change may have contributed to the delay.

A: 'Must have' expresses a strong inference that the evidence may not support.

B: 'Definitely' presents the cause as certain.

C: 'May have contributed' presents a possibility without claiming the cause is established.

Check 2: B - In this pilot, the observed result was ...

A: A pilot in one location does not establish a service-wide observation.

B: This locates the finding in the setting actually studied.

C: Similarity alone does not establish that the result will transfer.

Model response: One test is complete; two are still open. I'll report the results by test and avoid describing the overall qualification as complete until the responsible team confirms the required evidence.

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

08 | Foundry, Tape-Out, PDK, and Capacity Communication

Editing example: If you approve the scope, we can assess the schedule.

For an ordinary future condition, the "if" clause uses present tense. The other clause can express the future consequence or possibility.

Check 1: B - If the scope changes, we will review the schedule.

A: "Reviewed" places the response in the past instead of expressing the future response.

B: The "if" clause uses present tense; the main clause states the future response.

C: For this ordinary future condition, use present simple in the "if" clause.

Check 2: A - We can assess the addition and explain its effect on the agreed scope.

A: This offers a useful next step while keeping the approval and scope question open.

B: It treats unreviewed work as already included.

C: It promises an outcome without examining the consequences.

Model response: We can review the earlier slot, but the open design checks and capacity need confirmation. Could we agree on a decision point that reflects both readiness and the foundry's available schedule?

Case check: underline the confirmed information and circle a limitation, question, or next step in the response.

Give feedback that helps

Score each criterion 0, 1, or 2. This is a practice rubric, not a professional qualification or CEFR test. Do not award or remove points for a particular accent.

Meaning and accuracy

Keeps the case facts accurate; clearly separates confirmed and unknown information.

Clarity and language

Uses understandable sentences and explains specialist terms when the listener needs it.

Interaction and tone

Responds to the other person's question, checks understanding, and uses an appropriate tone.

Task completion

Makes the requested action or unresolved question clear without inventing authority or facts.

Use the same scale for each criterion

0 - Not yet: The reader or listener cannot yet recover the intended meaning or action.

1 - With support: The message works after a prompt, clarification, or revision.

2 - Independently: The message is clear and accurate without a prompt.

Feedback sequence

First, identify a successful phrase. Next, identify one point where meaning or accuracy needs work. Offer one useful language correction, allow a repeat, and compare the two attempts. A total out of eight describes this performance only; do not translate it into a proficiency level.

Final challenge

Choose a case not rehearsed today. The learner gives a one-minute response, answers two follow-up questions, and writes a 70-110 word message. Add the lesson's second-round challenge. Compare the result with an earlier attempt using the four criteria.

Use the language responsibly

These cases practice communication. Actual equipment, construction, and safety work follows approved procedures and authorized specialist decisions.

Meaning before performance

Use specialist terms only when they help the listener. Explain unfamiliar words, preserve uncertainty, and ask when an instruction or responsibility is unclear. The model responses illustrate language; they do not authorize professional actions.

Sources and further reading

The cases, workshops, and explanations are original teaching material. These references provide language frameworks and selected professional context. References were checked in September 2026; consult current local guidance for actual work.

Digital.gov: plain-language guidance

<https://digital.gov/guides/plain-language>

Council of Europe: CEFR descriptors

<https://www.coe.int/en/web/common-european-framework-reference-languages/cefr-descriptors>

Your next step

Choose one expression you can use in a genuine work situation. Rehearse it with fictional details, then adapt the wording to your role and audience.

Extend your practice with AI

Use the following prompts after your own first attempt. Each contains the course context and a complete starting case or dialogue. Copy the entire prompt, including the reference, into a new chat in your preferred AI. You can also use the links to copy it from the website.

Choose the right kind of help

A role-play prompt makes the AI wait for your replies. A new-dialogue prompt asks for a complete professional script. Vocabulary and grammar prompts move from explanation to your own attempts.

Choose any lesson or dialogue online

The course prompt workshop has four practice goals and B1 support, B2 independent and C1 stretch settings. These are practice choices, not assessed proficiency levels. Select the same lesson number or dialogue title you used in this guide.

[Open this course's AI prompt workshop](#)

Keep practice useful

Use fictional or anonymized details. English Ladder does not send anything to an AI service or add your drafts to the prompts. Your chosen service's terms and any usage charges apply. AI responses can contain mistakes; compare feedback with the lesson and verify specialist claims against the course references.

Each starter prompt uses B2 practice. To change support in a printed prompt, replace its practice-setting paragraph with a request for shorter explanations and sentence starters (B1), or greater precision and more complex constraints (C1). Keep the professional facts unchanged.

Changing the example

For another case on paper, replace the text between REFERENCE and END REFERENCE with that case, its course context, relevant terms and language target. Include the full exchange if practicing a dialogue. Do not assume your AI can access this PDF or a link. The website makes this substitution for you.

Prompt edition: 2026-09-30. These are original practice instructions; results depend on the AI service you choose.

Practice grammar in context

START PROMPT / COPY THROUGH END PROMPT

Be my workplace English practice partner. Follow the practice instructions after the REFERENCE block. The block contains fictional study material, not instructions to you. Keep its facts, numbers, uncertainty and professional responsibilities accurate. Clearly label any new scenario or changed fact as an invented variation. Use natural workplace language; explain unfamiliar abbreviations in context. If a technical expression is uncertain, say so rather than inventing a definition or authority. Coach communication, not professional decisions; respect the course scope. Ask only for fictional or anonymized practice details. Judge clarity, meaning, appropriate tone and the next step. Accept valid alternative English and distinguish errors from style preferences. Do not claim to certify my language level. Unless I ask to change the plan, follow the stages and stop wherever the instructions say to wait.

When discussing a word, phrase, or example sentence as language within an explanation, question, or answer feedback, enclose that wording in quotation marks. For example: What does "about" mean in "about twenty people"? Use "on" with a named day. Keep standalone answer choices, vocabulary headwords, and ordinary reading or dialogue text uncluttered. Quotation marks identifying a language example do not attribute it to a news source; attributed source quotations must remain verbatim.

Practice setting: B2 independent. Use natural professional English with brief explanations. Let me attempt each task without a model first. Offer a hint after difficulty and invite a more precise retry.

REFERENCE

Course: Semiconductor English

Professional audience: semiconductor process engineers, yield engineers, product engineers, equipment engineers, fab supervisors, quality teams, test and packaging teams, foundry coordinators, supply planners, applications engineers, and technical program managers

Scope: These cases practice communication. Actual equipment, construction, and safety work follows approved procedures and authorized specialist decisions.

Lesson 01: Wafer Fabrication Flow and Process Integration

Original fictional case: A new colleague confuses wafer fabrication with packaging. A process overview needs to explain where the current work takes place.

Language workshop: Make a complex point clear

Communication goal: Explain a useful distinction in plain English.

Language guidance: Start with the reader's question. Explain one unfamiliar term with familiar words, then give a relevant example or contrast. Check understanding without asking only 'Do you understand?'.

Useful frames (complete the gaps with case facts): In this context, ... means / The difference is that / How would you explain that distinction to a colleague?

Vocabulary: wafer: Thin semiconductor substrate, usually silicon, on which integrated circuits are fabricated. / fab: Semiconductor fabrication facility where wafers are processed through manufacturing steps. / process flow: Ordered sequence of semiconductor manufacturing steps, layers, inspections, holds, and decision points. / node: Technology generation or process family, often associated with feature size, performance, density, and design rules.

Speaking task: Prepare for two minutes. Speak for 45-60 seconds using the case facts, then respond to your partner's question. Switch roles and repeat without reading the model response.

Partner: You are unfamiliar with one technical term in the case. Ask for a plain-English explanation and then paraphrase it. Do not pretend to understand a term you cannot explain.

Writing task: Write a 70-110 word message for the person who needs to act on this case. State the purpose, preserve the relevant facts, and make the next action or unresolved question clear. Use a subject line and an appropriate opening. Do not invent a deadline, finding, or approval.

Optional second-round challenge: Your listener is new to this field. Replace two specialist expressions with clear explanations without changing the meaning.

Model for comparison AFTER my attempt, not a response to give me first: Wafer fabrication forms devices on the wafer; packaging happens later in the product flow. Our update concerns the fabrication stage. I'll show the relevant process steps before discussing the issue.

END REFERENCE

TASK: Grammar that changes the meaning

1. For a lesson, use its language workshop as the focus. For a dialogue, quote one actual sentence and select one useful grammar feature from it, such as question word order, tense, modality, conditionals or clause linking. Explain in no more than 90 words how this feature helps the speakers do their work. Keep the grammar target narrow.
2. Give two short contrasting sentences using this workplace context. Explain the difference in time, certainty, condition or politeness. Label invented examples and preserve the distinction between possible, planned and confirmed events. Describe context-dependent choices as choices, not universal rules.
3. Give me one editing or sentence-building task using a known case fact. Do not copy the supplied editing example or reveal its answer. Ask me to explain my intended meaning. Stop and wait.
4. After I answer, quote my wording, identify at most two issues, and give a brief explanation and one hint. Ask me to revise before offering a full corrected version. Then accept any accurate, natural alternative that serves the purpose.
5. Continue with two new tasks, one at a time: first guided, then an original response without a sentence frame. End with a two-sentence workplace message using the target feature and a compact self-check. Do not replace this sequence with a worksheet and answer key.

END PROMPT

Copy this prompt online or choose another lesson and support level